Practical No. 11 : Study & Implementation of association rule mining.

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Roll No.: 75

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
import pandas as pd
from mlxtend.frequent_patterns import apriori, association_rules


data = {
    'Milk':      [1, 1, 0, 1, 0, 1],
    'Bread':     [1, 1, 1, 1, 0, 1],
    'Butter':    [0, 1, 1, 1, 0, 1],
    'Biscuits': [0, 0, 1, 0, 1, 0]
}

df = pd.DataFrame(data)


frequent_items = apriori(
    df.astype(bool),
    min_support=0.3,
    use_colnames=True
)

print("Frequent Itemsets:")
print(frequent_items)


rules = association_rules(
    frequent_items,
    metric="confidence",
    min_threshold=0.6
)

print("\nAssociation Rules:")
print(rules[['antecedents', 'consequents',
             'support', 'confidence', 'lift']])
```

Frequent Itemsets:

	support	itemsets
0	0.666667	frozenset({Milk})
1	0.833333	frozenset({Bread})
2	0.666667	frozenset({Butter})
3	0.333333	frozenset({Biscuits})
4	0.666667	frozenset({Milk, Bread})
5	0.500000	frozenset({Milk, Butter})
6	0.666667	frozenset({Bread, Butter})
7	0.500000	frozenset({Milk, Bread, Butter})

Association Rules:

	antecedents	consequents	support	\
0	frozenset({Milk})	frozenset({Bread})	0.666667	
1	frozenset({Bread})	frozenset({Milk})	0.666667	
2	frozenset({Milk})	frozenset({Butter})	0.500000	
3	frozenset({Butter})	frozenset({Milk})	0.500000	
4	frozenset({Bread})	frozenset({Butter})	0.666667	
5	frozenset({Butter})	frozenset({Bread})	0.666667	
6	frozenset({Milk, Bread})	frozenset({Butter})	0.500000	
7	frozenset({Milk, Butter})	frozenset({Bread})	0.500000	
8	frozenset({Bread, Butter})	frozenset({Milk})	0.500000	
9	frozenset({Milk})	frozenset({Bread, Butter})	0.500000	
10	frozenset({Bread})	frozenset({Milk, Butter})	0.500000	
11	frozenset({Butter})	frozenset({Milk, Bread})	0.500000	

	confidence	lift
0	1.00	1.200
1	0.80	1.200
2	0.75	1.125
3	0.75	1.125
4	0.80	1.200
5	1.00	1.200
6	0.75	1.125
7	1.00	1.200
8	0.75	1.125
9	0.75	1.125
10	0.60	1.200
11	0.75	1.125